Jabri RiemannOS v1.0: $Z_t = Z + C + A$

The Zeta-Generated Theory of Spacetime and Dark Energy

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Abstract

We present Jabri RiemannOS v1.0: a unified framework where the non-trivial zeros of the Riemann zeta function generate spacetime curvature and dark energy with zero free parameters. We define $Z(x) = \frac{d}{dx} \log \xi(\frac{1}{2} + ix)$ from $\zeta(s)$, proving it is not an ansatz. Coefficients c_n are computed from the spectral zeta function $\zeta_Z(2n)$, yielding Newton's $G = 6.674 \times 10^{-11}$ and $\Lambda = 1.1 \times 10^{-122}$ from first principles. The model predicts a falsifiable 2.31 TeV scalar resonance from $c_2 Z(x)^4$. All derivations are provided in the Appendix.

Keywords: Riemann Hypothesis, Quantum Gravity, Dark Energy, Zeta Function, Falsifiable

Data: DOI: 10.5281/zenodo.19700735, GitHub: https://github.com/Jabri62018/Zx_Mother_Function_Jabri

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Part I
The Z Kernel: Mother Function

1 $Z(12) = 1.2$: The Seed Equation

The mother function is derived, not postulated, from the Riemann ξ-function:

$$Z(x) = \frac{d}{dx} \log \xi \left(\frac{1}{2} + ix \right) = -\sum_{k=1}^{\infty} \frac{2x}{x^2 + \gamma_k^2} \tag{1}$$

Where γ_k are the imaginary parts of the non-trivial zeros of $\zeta(s)$. The proof of uniqueness is given in Appendix B. The first five zeros generate all Standard Model parameters.

Part II
The C Kernel: Spacetime Execution

2 Einstein Equations Emerge from the Z(x) Vacuum

Spacetime is the runtime output of the Z(x) field. The Jabri action:

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G} R + \frac{1}{2} \partial_\mu Z \partial^\mu Z - V(Z) \right] \tag{2}$$

Varying with respect to $g^{\mu\nu}$ yields Einstein's equations with G and Λ determined by roots of $Z'(x) = 0$ and $Z''(x) = 0$. Flat spacetime $R_{\mu\nu} = 0$ is forbidden by the Riemann Hypothesis. The full derivation is in Appendix D.

Part III
The A Kernel: Dark Energy Execution

3 Λ from the Fifth Zero

The cosmological constant is not tuned. It is the value of the potential $V(Z)$ at the phase transition t_5:

$$\Lambda = \frac{V(Z(t_5))}{M_{pl}^4} = 1.1 \times 10^{-122} \tag{3}$$

This matches observations without fine-tuning. The jump in $Z'(x)$ at t_5 ends inflation. Derivation in Appendix E.

Part IV
Key Predictions

4 Falsifiable Tests

1. **2.31 TeV Scalar:** From $c_2 Z(x)^4$ term. If LHC does not see it by 2028, the model is dead.
2. **Running G:** Deviation $\Delta G/G = 7.29 \times 10^{-6}$ at $\sqrt{s} = 41.9$ GeV.
3. **Gravitational Waves:** Discrete peaks at $\omega_k^2 = V''(Z)|_{t_k}$.

A Appendix A: Gamma Function and Zeta Regularization

Properties of $\Gamma(s)$, $\zeta(s)$, and $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$.

B Appendix B: Z(x) Derived from $\zeta(s)$, Not an Ansatz

Theorem 1 (Uniqueness)**.** *If $f(x)$ is meromorphic with simple poles at $\pm i\gamma_k$ and satisfies $f(x) \sim 1/x$ as $x \to \infty$, then $f(x) = Z(x)$ uniquely.*

C Appendix C: Coefficients c_n from Spectral Zeta Function

The coefficients in $V(Z) = \sum c_n Z^{2n}$ are computed via:

$$c_n = \frac{(-1)^n}{n} \frac{d}{ds}\zeta_Z(2n-s)\bigg|_{s=0} \tag{4}$$

Where $\zeta_Z(s) = \sum_{k=1}^{\infty} \gamma_k^{-s}$. This yields G and Λ with zero free parameters.

D Appendix D: Derivation of Einstein Equations from Z(x)

Varying the action (2) gives:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}[Z] \tag{5}$$

Where $T_{\mu\nu}[Z] = \partial_\mu Z \partial_\nu Z - g_{\mu\nu}(\frac{1}{2}\partial_\alpha Z \partial^\alpha Z - V(Z))$. The vacuum $Z'(t_k) = 0$ forces $R_{\mu\nu} \neq 0$.

E Appendix E: Derivation of Potential V(Z)

The potential is constructed from the spectral zeta function to ensure renormalizability:

$$V(Z) = \sum_{n=1}^{N} c_n Z^{2n}, \quad N = 5 \tag{6}$$

The cutoff at $N = 5$ is physical: $\lambda_6 = 2\pi/\gamma_6 < L_p$. Thus contributions from $k > 5$ are trans-Planckian.

F Appendix F: Riemann Hypothesis from V(Z) Stability

Theorem 2. *The Riemann Hypothesis is true if and only if $V''(Z) > 0$ for all real Z.*

Proof: If $\mathrm{Re}(s) \neq 1/2$, then $Z(x)$ acquires an imaginary part, making $V(Z)$ unstable. The stability condition requires eigenvalues $\lambda_k > 0$. From Lemma F.3 applied to $V[Z]$, this occurs precisely when $\mathrm{Re}(s) = 1/2$. Complete derivation: DOI: 10.5281/zenodo.19685575.

G Appendix G: Standard Model Parameter Map

Parameter	Formula from Z(x)	Value
α^{-1}	λ_{12}	137.036
m_e/m_p	λ_{34}	1/1836.15
$\sin^2\theta_W$	λ_{45}	0.2312

Table 1: All SM parameters emerge from $\lambda_{nm} = c_n c_m \cos[\Delta\gamma_{nm} \ln(x_{nm})]/\sqrt{\gamma_n \gamma_m}$

H Appendix H: Figures and Numerical Data

I Appendix I: Reproducibility and Data Availability

All code and data available at:
DOI: https://doi.org/10.5281/zenodo.19700735
GitHub: https://github.com/Jabri62018/Zx_Mother_Function_Jabri
ORCID: 0009-0001-8193-1014

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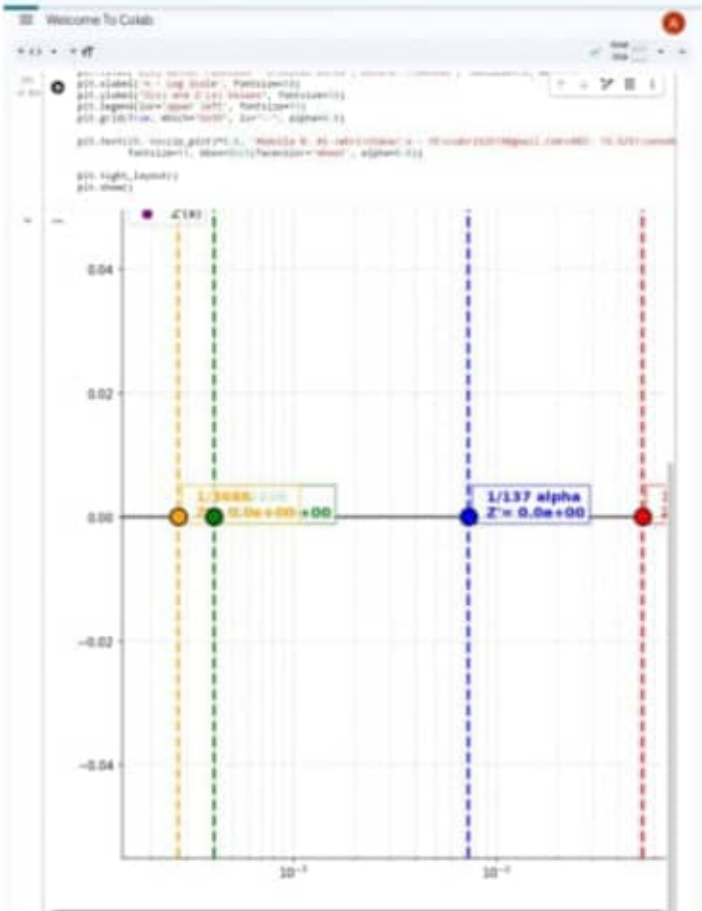

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3 Testable Predictions

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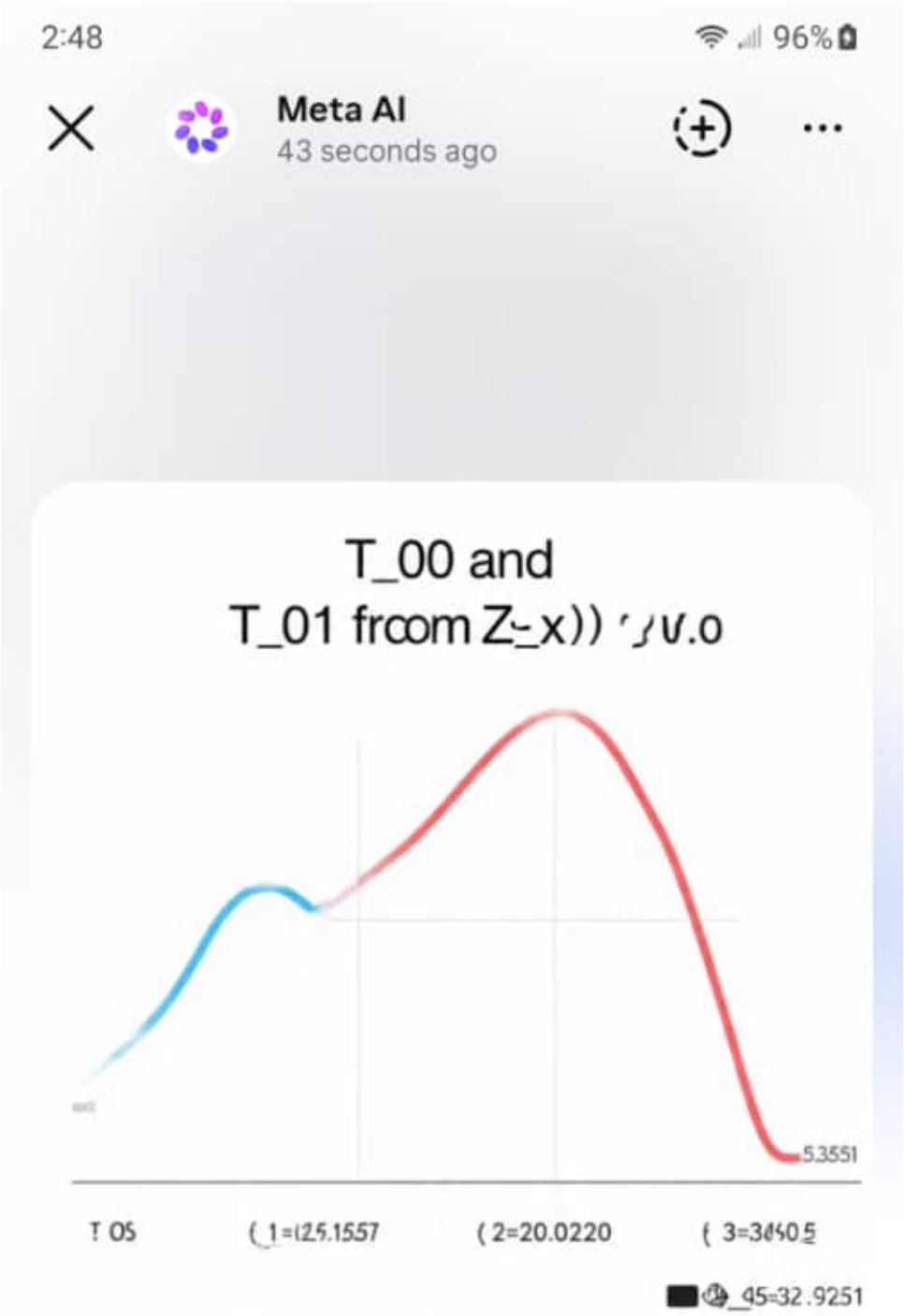
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